

Right multiplication scales every maximal minor by one determinant

MathlibAnnex.Matrix.maximalMinors_mul

theorem

Turns a square right factor into a common scalar on the family of maximal minors.

Statement

For an I -by- n matrix A and an n -by- n matrix L over a commutative ring, with I linearly ordered, write $\mathcal{M}(A)$ for the family of determinants of all n -row submatrices in increasing row order. Then

$$\mathcal{M}(AL) = (\det L) \mathcal{M}(A).$$

The equality is pointwise on n -element row subsets, using increasing row order on both sides.

Assumptions

The integer n may be zero; the row set may be infinite. The coefficient ring is commutative. The square factor L need not be invertible.

Conclusion

For each selected subset S , $\det((AL)_S) = (\det L) \det(A_S)$. Thus one scalar, independent of S , scales the whole family.

Related identities in the same source distinguish scalar multiplication from right multiplication: $\mathcal{M}(cA) = c^n \mathcal{M}(A)$. The zero-matrix family vanishes when $n > 0$; this restriction matters, because the empty determinant for $n = 0$ is 1. Selection commutes also with application to a column vector: $(Av)|_S = A_S v$. These are separate identities, not invertibility assumptions for the displayed theorem.

Proof route

Select the same rows before and after multiplication, then apply multiplicativity of the square determinant.

▼ Proof steps

1. For an n -element subset S , matrix multiplication is a sum over the finite column set. Hence selecting rows commutes with that sum: $(AL)_S = A_S L$.
2. Both selected matrices are square over a commutative ring, so determinant multiplicativity applies:

$$\mathcal{M}(AL)(S) = \det(A_S L) = \det(A_S) \det L = (\det L) \mathcal{M}(A)(S).$$

Commutativity gives the scalar in the stated order. Since this holds for every S , the two functions are equal. For $n = 0$ the same calculation reads $1 = 1 \cdot 1$.

Main citations

- Exact declaration and its source — `MathlibAnnex.Matrix.maximalMinors_mul`
- The component minor multiplication formula — `MathlibAnnex.Matrix.maximalMinor_mul`
- Scalar multiplication scales minors by the dimension power — `MathlibAnnex.Matrix.maximalMinors_smul`
- Selecting the zero matrix — `MathlibAnnex.Matrix.maximalSubmatrix_zero`
- A zero minor in positive dimension — `MathlibAnnex.Matrix.maximalMinor_zero_of_pos`
- The zero family in positive dimension — `MathlibAnnex.Matrix.maximalMinors_zero_of_pos`
- Selection commutes with a matrix-vector product — `MathlibAnnex.Matrix.maximalSubmatrix_mulVec`

Lean source signature (exact)

```
theorem maximalMinors_mul
  {n : ℕ} {ι : Type u} {R : Type v} [LinearOrder ι] [CommRing R]
  (A : _root_.Matrix ι (Fin n) R)
  (L : _root_.Matrix (Fin n) (Fin n) R) :
  maximalMinors (A * L) = L.det • maximalMinors A
```

In the source	Mathematical meaning
$\{n : \mathbb{N}\} \{\iota : \text{Type } u\} [\text{LinearOrder } \iota] \{R : \text{Type } v\} [\text{CommRing } R]$	$n \geq 0$, an ordered row set I and a commutative ring R ; I need not be finite.
$(A : \text{root}.Matrix \iota (\text{Fin } n) R)$	The I -by- n matrix A .
$(L : \text{root}.Matrix (\text{Fin } n) (\text{Fin } n) R)$	The n -by- n right factor L , with no invertibility assumption.
$\text{maximalMinors } (A * L)$	The family $\mathcal{M}(AL)$; its S component is $\det((AL)_S)$ with rows of S taken increasingly.

In the source	Mathematical meaning
$L.\text{det} \bullet \text{maximalMinors } A$	The pointwise scalar multiple $(\det L)\mathcal{M}(A)$, whose S component is $(\det L) \det A_S$.
$\text{maximalMinors } (A * L) = L.\text{det} \bullet \text{maximalMinors } A$	Equality of the two families at every n -element row subset: $\det((AL)_S) = (\det L) \det A_S$.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section

open scoped BigOperators
```

```
namespace MathlibAnnex
namespace Matrix

universe u v
```

▼ **Exact content identity**

Declaration: `MathlibAnnex.Matrix.maximalMinors_mul`

Accepted content SHA-256: 9916875d513ea3e76b5ac1edecfe40877bd5507f8995d8f9d5bef680d2162599

Accepted source guide SHA-256: f363e4d8026edadd112bd44cc8a2453a44d1ae7b27e265816c59cac58317124d

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73